



Renewals Methodology

12/4/21 - Ryan's attempt to build a standardized methodology ('Renewals' dataset in Sigma)

- For each year a user has at least one paid subscription satisfying the following properties:
 - Billing frequency = 12
 - Payment provider != admin, spouse, gift_sub or null
 - Name = Qeepsake Premium, Qeepsake Plus, or Qeepsake Standard
- Take the **latest** of those subscriptions (this is what we consider as the “real subscription” since it sets the expected renewal date)
- If the year in question **isn't the latest year that we have such an object for that user (AKA, if we see that the user started on a paid subscription in at least one future year), mark this subscription as “Renewed”**
- Take the date difference between the expected renewal date (the date of the subscription + 365 days) and the next user subscription we see for that user.
 - This is the **day of renewal** (ie, how many days did it take the user to renew?) For a typical auto-renewal, this will be 0. In some cases it will be a large positive integer, and in some very rare cases it will be negative, if the user proactively bought a new paid sub earlier than they came up for renewal next year.
- Append any properties about the user and that year's last subscription (the subscription for which we're asking the question, “did the user renew this subscription in a following year?”) that we wish to filter by

- The **renewal rate for any particular time window** [filtered by any particular user/subscription properties] is the % of subscriptions we see in that time period that we see have already been renewed in a subsequent year. Furthermore, we can define an “X-day renewal rate” by filtering down to day of renewal $\leq X$. Our standard window will be a 30-day renewal rate as Jeff & I agreed upon.
 - Note that this definition is technically looking at 1 year back from when the renewals happen. Our renewal rate “last month” will be the “renewal rate” for subscriptions that started that month in the previous year

Ex. #1: In December 2021, we will report the **November 2020 Renewal Rate**, which is the % of November 2020 Subscriptions that we can already see a renewal for as of the current day. Practically, this will be near maturity on Dec 1, and we can snapshot it to just 30-day renewals if we want to be able to report full maturity to compare with previous months in 2020.

Ex. #2: **2018’s Qeepsake Premium Renewal Rate** is the % of **2018 Premium Subscriptions** where we see **any** new sub in a later year for that user as of the current day (and were that user’s last subscription in 2018).

Ex. #3: The **2019 IOS Year 2 30-day renewal rate** is the % of **2019 IOS Subscriptions** that...

- Belong to a user who first subscribed in 2018 (on any payment provider)
- Are the last subscription we saw for that user in 2019

...For which we saw a renewal within 30 days of the expected 2020 renewal date.

Press 'space' for AI, '/' for commands...

This was a followup analysis I did to ensure that the methodology above was a roughly accurate representation of what was actually happening to subscriptions in our userbase. Leaving it here in case it's useful but feel free to ignore.

12/16/21 - Triple-checking renewals methodology (see [DATA-147](#))

Checking denominators:

- Total User Subscriptions in our DB: 2.7M

- Total with Billing Frequency 12: 1.9M
 - Total paid subs with non-12 billing frequency: 8.5K
- Total Annual User Sub with non-null payment provider: 367K
- Total Annual User Sub with "paid name" and non-null payment provider: 265K
 - There are 102K subs with non-null payment provider and name = Qeepsake Lite. Almost all (101K+) have source = "renewed on Lite plan" or source = "Renewed existing subscription" or source = "Active Admin Created new user subscription"
- Total Annual User Sub with "paid name" and payment provider = "stripe", "ios_iap" or "stripe_android" (**AKA those being used in this analysis**): 239K
 - Of the 26K with non-standard payment providers, 14.3K are spouse, 5.8K are admin, 1.8K are gift, and 15K are "null". Of the Nulls:
 - 4.3K are accounted for as gift subs
 - 1.5K are accounted for by the share offer
 - 8.5K have source = null and status = revoked. 10+ spot checks identify these as users who have hard-deleted their accounts (*****s in Active Admin)

Conclusion: The user subscriptions being counted in this renewals analysis make up 85% of paid subscriptions and 90% of paid annual subscriptions. The paid annual subscriptions we are excluding are all corner cases that we intend to exclude: Spouses, Gift Subs, Admin Subs, and Deleted Users.

Investigating renewals happening 14 days or more outside the expected end date:

To get specific on the numbers, by this methodology we see ~6K instances where a successful renewal happened 15 or more days after the "expected renewal date", out of 112K total successful renewals.

- The main explanation for this is simply that the current methodology doesn't exclude it from happening at all! Because we're looking at any and all paid subscriptions that happen in the years following a user's last one, "successful renewals" don't have to be tied to a successful code-based renewal transaction at all. They can simply happen when a user fails to renew, but stays on Qeepsake Lite... then some time later upgrades again to a paid plan.
- As for the rare cases where we attempt to renew users more than 1Y after their subscription:
 - We have <100 active subs with "subscription ends on" more than 1Y after their start date. About half are admin-created or spouses, the rest are mainly gift subs and upgrades. I don't think this should be a concern, either.

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